Ira Geller

(443) 608-4601; ibg24@cornell.edu

Education

Cornell University, College of Engineering; Expected Graduation: December 2027; GPA: 4.0
- Bachelor of Science in Mechanical Engineering
- Relevant courses: Fluid Mechanics, System Dynamics, Mechanics of Materials, Statics, Thermodynamics

Engineering Experience

Northrop Grumman (Active DoD Secret Security Clearance)

Structural Analysis Engineering Intern — *Summer 2026*
- Redesigned an NX electronics cover to achieve positive fatigue margins under increased vibration loads
- Built Siemens NX CAE finite element models for airborne sensor electronics to support design reviews
- Performed hand calculations for bolted joints, structural strength, and margins of safety to verify flight hardware
- Collaborated with thermal engineers to reduce heatsink mass while meeting structural and thermal requirements
- Produced manufacturing-ready drawings using GD&T and applicable industry standards

Mechanical Engineering Intern — *Summer 2025*
- Designed and ran Instron compression tests on Megtron 7 to establish allowables for high-load designs
- Planned and executed random-vibration testing to compare thermal interface material performance
- Reviewed and validated material properties for more than 100 commonly used materials

Systems Engineering Intern — *Summer 2024*
- Led the development of an automatic calibration sequence for an F-22 test bench
- Built a MATLAB interface to operate the automated calibration sequence

Cornell Design Build Fly

Chief Engineer / Team Lead — *2026–Present*
- Provide technical direction and oversee daily operations for a 40-member team with a $40,000 budget
- Lead clean-sheet design, fabrication, testing, failure analysis, and iteration of multiple aircraft prototypes
- Coordinate aerodynamic, structural, propulsion, and electrical interfaces through design reviews
- Develop a Python-based aircraft sizing and optimization tool for a 31-variable design space

Mechanical and Structural Subteam Lead / Integration Lead — *2025–2026*
- Led a 10-member subteam through aircraft design, first-principles analysis, design reviews, prototype fabrication, and flight testing
- Primary author of a 60-page technical report covering requirements, design, analysis, and testing
- Developed carbon-fiber wet-layup and vacuum-bagging processes, improving repeatability and part quality
- Designed and fabricated a carbon-fiber fuselage and 3D-printed nose cone, and integrated electronics packaging
- Mentored five new team members in aircraft design theory, CAD skills, and aircraft fabrication skills

Mechanical and Structural Subteam Member — 2024–2025
- Led the design, integration, and testing of a glider release mechanism
- Built ANSYS static structural models to evaluate loads on the aircraft's monocoque fuselage

Leadership & Software Experience

Cornell Big Red Sports Network

Club Vice President — *2026–Present*
- Coordinate operations across seven departments within a 100-person sports media organization

Vice President of Analytics — *2025–2026*
- Directed software development for a 35-person department providing analytics for three Division I Cornell sports
- Architected and led the development of a database-driven pitching analytics website – brsnanalytics.com
- Led department-wide training in data analytics practices and project workflows

Baseball Analyst, Analytics Department — *2024–2025*
- Pioneered the automation of batter data collection for baseball reports using Python and Google APIs
- Produced weekly scouting reports identifying the strengths and weaknesses of opposing batters

Skills and Interests

Skills: CAD/PLM: Siemens NX, Teamcenter, SolidWorks; Analysis: NX CAE, ANSYS FEA, ANSYS Fluent (CFD); Manufacturing: Machining (mill/lathe), 3D printing, injection molding, carbon-fiber composites, DFMA; Programming: Python, MATLAB
Interests: Rockets, Board Games, Cooking + Baking, Survivor (the TV show)